

CHAPTER 3

3. MEASURES OF CENTRAL TENDENCY

Introduction

- When we want to make comparison between groups of numbers it is good to have a single value that is considered to be a good representative of each group. This single value is called the **average** of the group. Averages are also called measures of central tendency.
- Measures of central tendency are numerical measures which intends to describe the middle value or the central value or the typical value in a given data set.
- An average which is representative is called typical average and an average which is not representative and has only a theoretical value is called a descriptive average. A typical average should possess the following:
 - It should be rigidly defined, exist and be unique.
 - It should be based on all observation under investigation.
 - It should be as little as affected by extreme observations.
 - It should be capable of further algebraic treatment.
 - It should be as little as affected by fluctuations of sampling.
 - It should be ease to calculate and simple to understand.

Objectives:

- ☞ To comprehend the data easily or to describe a data in a concise manner.
- ☞ To facilitate comparison.
- ☞ To summarizing/reducing the volume of the data
- ☞ To make further statistical analysis.

The Summation Notation:

- Let $X_1, X_2, X_3 \dots X_N$ be a number of measurements where N is the total number of observation and X_i is i^{th} observation.
- Very often in statistics an algebraic expression of the form $X_1 + X_2 + X_3 + \dots + X_N$ is used in a formula to compute a statistic. It is tedious to write an expression like this very often, so mathematicians have developed a shorthand notation to represent a sum of scores, called the summation notation.
- The symbol $\sum_{i=1}^N X_i$ is a mathematical shorthand for $X_1 + X_2 + X_3 + \dots + X_N$

$$\sum_{i=1}^N X_i = X_1 + X_2 + \dots + X_N$$

The expression is read, "the sum of X sub i from i equals 1 to N ." It means "add up all the numbers."

Example: Suppose the following were scores made on the first homework assignment for five students in the class: 5, 7, 7, 6, and 8. In this example set of five numbers, where $N=5$, the summation could be written:

$$\sum_{i=1}^5 X_i = X_1 + X_2 + X_3 + X_4 + X_5 = 5 + 7 + 7 + 6 + 8 = 33$$

The "i=1" in the bottom of the summation notation tells where to begin the sequence of summation. If the expression were written with "i=3", the summation would start with the third number in the set. For example:

$$\sum_{i=3}^N X_i = X_3 + X_4 + \dots + X_N$$

In the example set of numbers, this would give the following result:

$$\sum_{i=3}^N X_i = X_3 + X_4 + X_5 = 7 + 6 + 8 = 21$$

The "N" in the upper part of the summation notation tells where to end the sequence of summation. If there were only three scores then the summation and example would be:

$$\sum_{i=1}^3 X_i = X_1 + X_2 + X_3 = 5 + 7 + 7 = 21$$

Sometimes if the summation notation is used in an expression and the expression must be written a number of times, as in a proof, then a shorthand notation for the shorthand notation is employed. When the summation sign $\sum$ is used without additional notation, then "i=1" and "N" are assumed.

$$\sum X = \sum_{i=1}^N X_i = X_1 + X_2 + \dots + X_N$$

For example:

PROPERTIES OF SUMMATION

1. $\sum_{i=1}^n k = nk$ where k is any constant
2. $\sum_{i=1}^n kX_i = k \sum_{i=1}^n X_i$ where k is any constant
3. $\sum_{i=1}^n (a + bX_i) = na + b \sum_{i=1}^n X_i$ where a and b are any constant
4. $\sum_{i=1}^n (X_i + Y_i) = \sum_{i=1}^n X_i + \sum_{i=1}^n Y_i$
5. $\sum_{i=1}^n x_i y_i \neq \sum_{i=1}^n x_i \sum_{i=1}^n y_i$

The sum of the product of the two variables could be written:

$$\sum_{i=1}^N (X_i * Y_i) = (X_1 * Y_1) + (X_2 * Y_2) + \dots + (X_N * Y_N)$$

Example: considering the following data determine

X	Y
5	6
7	7

7	8
6	7
8	8

a). $\sum_{i=1}^5 X_i$

b). $\sum_{i=1}^5 Y_i$

c). $\sum_{i=1}^5 10$

d). $\sum_{i=1}^5 (X_i + Y_i)$

e). $\sum_{i=1}^5 (X_i - Y_i)$

f). $\sum_{i=1}^5 X_i Y_i$

g). $\sum_{i=1}^5 X_i^2$

h). $(\sum_{i=1}^5 X_i)(\sum_{i=1}^5 Y_i)$

Solutions:

a). $\sum_{i=1}^5 X_i = 5 + 7 + 7 + 6 + 8 = 33$

b). $\sum_{i=1}^5 Y_i = 6 + 7 + 8 + 7 + 8 = 36$

c). $\sum_{i=1}^5 10 = 5 * 10 = 50$

d). $\sum_{i=1}^5 (X_i + Y_i) = (5 + 6) + (7 + 7) + (7 + 8) + (6 + 7) + (8 + 8) = 69 = 33 + 36$

e). $\sum_{i=1}^5 (X_i - Y_i) = (5 - 6) + (7 - 7) + (7 - 8) + (6 - 7) + (8 - 8) = -3 = 33 - 36$

f). $\sum_{i=1}^5 X_i Y_i = 5 * 6 + 7 * 7 + 7 * 8 + 6 * 7 + 8 * 8 = 241$

g). $\sum_{i=1}^5 X_i^2 = 5^2 + 7^2 + 7^2 + 6^2 + 8^2 = 223$

h). $(\sum_{i=1}^5 X_i)(\sum_{i=1}^5 Y_i) = 33 * 36 = 1188$

Types of measures of central tendency

There are several different measures of central tendency; each has its advantage and disadvantage.

- The Mean (Arithmetic, Geometric and Harmonic)
- The Mode
- The Median
- Quantiles (Quartiles, Deciles and Percentiles)

The choice of these averages depends on which best fit the property under discussion.

The Arithmetic Mean

- Is defined as the sum of the magnitude of the items divided by the number of items.
- The mean of $X_1, X_2, X_3 \dots X_n$ is denoted by A.M ,m or $\bar{X}$ and is given by:

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} \Rightarrow \bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

The **population mean**, μ and is calculated as follows:

$$\mu = \frac{x_1 + x_2 + x_3 + \dots + x_N}{N} = \frac{\sum x}{N}, \text{ N is the number of values in the population.}$$

But

- If X_1 occurs f_1 times
- If X_2 occurs f_2 times
- .
- .
- If X_n occurs f_n times

Then the mean will be $\bar{X} = \frac{\sum_{i=1}^k f_i X_i}{\sum_{i=1}^k f_i}$, where k is the number of classes and $\sum_{i=1}^k f_i = n$

Example: Obtain the mean of the following number

2, 7, 8, 2, 7, 3, 7

Solution:

X_i	f_i	$X_i f_i$
2	2	4
3	1	3
7	3	21
8	1	8
Total	7	36

$$\bar{X} = \frac{\sum_{i=1}^5 X_i}{n} = \frac{158 + 189 + 265 + 127 + 191}{5} = \frac{930}{5} = 186$$

Short cut Method:

- ✓ Subtract an constant "c" from each observation, $X' = X - c$

- ✓ Calculate the mean of the transformed data $\bar{X}' = \frac{\sum_{i=1}^5 X'_i}{n}$

- ✓ Then $\bar{X} = \bar{X}' + c$

Use the above data to calculate the mean in a short cut method .let $c=100$

Example: the following data give the price of 5 houses sold recently in Addis in thousands of Birr.

158, 189, 265, 127, 191: find the mean sales price for these houses.

$$\bar{X} = \frac{\sum_{i=1}^4 f_i X_i}{\sum_{i=1}^4 f_i} = \frac{36}{7} = 5.15$$

Arithmetic Mean for Grouped Data

If data are given in the shape of a continuous frequency distribution, then the mean is obtained as follows:

$$\bar{X} = \frac{\sum_{i=1}^k f_i X_i}{\sum_{i=1}^k f_i}, \text{ Where } X_i = \text{the class mark of the } i^{\text{th}} \text{ class and } f_i = \text{the frequency of the } i^{\text{th}} \text{ class}$$

Example: calculate the mean for the following age distribution.

Class	frequency
6- 10	35
11- 15	23
16- 20	15
21- 25	12
26- 30	9
31- 35	6

Solutions:

- First find the class marks
- Find the product of frequency and class marks
- Find mean using the formula.

Class	f_i	X_i	$X_i f_i$
6- 10	35	8	280
11- 15	23	13	299
16- 20	15	18	270
21- 25	12	23	276
26- 30	9	28	252
31- 35	6	33	198
Total	100		1575

$$\bar{X} = \frac{\sum_{i=1}^6 f_i X_i}{\sum_{i=1}^6 f_i} = \frac{1575}{100} = 15.75$$

Short cut method

- ✓ Subtract any constant “c” from the class mark and divided by the class width to obtained a transformed midpoint, M' as $M' = m - c/w$

- ✓ Calculate the mean of the transformed data/distribution, $\bar{x}' = \sum \frac{M'_i f_i}{n}$
- ✓ Then the mean of the original data is $\bar{x} = \bar{x}' \times w + c$

Exercises:

1. Marks of 75 students are summarized in the following frequency distribution:

Marks	No. of students
40-44	7
45-49	10
50-54	22
55-59	f_4
60-64	f_5
65-69	6
70-74	3

If 20% of the students have marks between 55 and 59

- i. Find the missing frequencies f_4 and f_5 .
 - ii. Find the mean.
- If the values in a series or mid values of a class are large enough, coding of values is a good device to simplify the calculations.
 - For raw data suppose we have used the following coding system.

$$d_i = X_i - A$$

$$\Rightarrow X_i = d_i + A, \bar{X} = \frac{\sum_{i=1}^n X_i}{n} = \frac{\sum_{i=1}^n (d_i + A)}{n} \Rightarrow \bar{X} = A + \frac{\sum_{i=1}^n d_i}{n} \Rightarrow \bar{X} = A + \bar{d}$$

Where A is an assumed mean and $\bar{d}$ is the mean of the coded data.

- If the data are expressed in terms of ungrouped frequency distribution

$$d_i = X_i - A$$

$$\Rightarrow X_i = d_i + A, \bar{X} = \frac{\sum_{i=1}^k f_i X_i}{n} = \frac{\sum_{i=1}^k f_i (d_i + A)}{n} \Rightarrow \bar{X} = A + \frac{\sum_{i=1}^k f_i d_i}{n} \Rightarrow \bar{X} = A + \bar{d}$$

- In both cases the true mean is the assumed mean plus the average of the deviations from the assumed mean.
- Suppose the data is given in the shape of continuous frequency distribution with a constant class size of w then the following coding is appropriate.

$$d_i = \frac{X_i - A}{w} \Rightarrow X_i = w d_i + A, \bar{X} = \frac{\sum_{i=1}^k f_i X_i}{n} = \frac{\sum_{i=1}^k f_i (w d_i + A)}{n}$$

$$\Rightarrow \bar{X} = A + \frac{\sum_{i=1}^k f_i w d_i}{n} \Rightarrow \bar{X} = A + w \bar{d}$$

Where: X_i is the original class mark for the i^{th} class.

d_i is the transformed class mark for the i^{th} class.

A is an assumed mean usually the mean of the class marks.

($i = 1, 2, \dots, k$)

Example:

1. Suppose the deviations of the observations from an assumed mean of 7 are: 1, -1, -2, -2, 0, -3, -2, 2, 0, -3.

- a) Find the true mean
- b) Find the original observation.

Solutions:

$$A = 7, \sum_{i=1}^{10} d_i = -10$$

$$\text{a) } \Rightarrow \bar{d} = \frac{-10}{10} = -1$$

$$\Rightarrow \bar{X} = A + \bar{d} = 7 - 1 = 6$$

The true mean is 6.

- b) Using $X_i = A + d_i$ we obtain the following original observations:
8, 6, 5, 5, 7, 4, 5, 9, 7, 4.

Special properties of Arithmetic mean

1. The sum of the deviations of a set of items from their mean is always zero. i.e.

$$\sum_{i=1}^n (X_i - \bar{X}) = 0.$$

2. The sum of the squared deviations of a set of items from their mean is the minimum. i.e.

$$\sum_{i=1}^n (X_i - \bar{X})^2 < \sum_{i=1}^n (X_i - A)^2, A \neq \bar{X}$$

3. If $\bar{X}_1$ is the mean of n_1 observations

If $\bar{X}_2$ is the mean of n_2 observations

.

.

If $\bar{X}_k$ is the mean of n_k observations

Then the mean of all the observation in all groups often called the combined mean is given by:

$$\bar{X}_c = \frac{\bar{X}_1 n_1 + \bar{X}_2 n_2 + \dots + \bar{X}_k n_k}{n_1 + n_2 + \dots n_k} = \frac{\sum_{i=1}^k \bar{X}_i n_i}{\sum_{i=1}^k n_i}$$

Example: In a class there are 30 females and 70 males. If females averaged 60 in an examination and boys averaged 72, find the mean for the entire class.

Solutions:

Females

$$\bar{X}_1 = 60$$

$$n_1 = 30$$

Males

$$\bar{X}_2 = 72$$

$$n_2 = 70$$

$$\bar{X}_c = \frac{\bar{X}_1 n_1 + \bar{X}_2 n_2}{n_1 + n_2} = \frac{\sum_{i=1}^2 \bar{X}_i n_i}{\sum_{i=1}^2 n_i} \Rightarrow \bar{X}_c = \frac{30(60) + 70(72)}{30 + 70} = \frac{6840}{100} = 68.40$$

4. If a wrong figure has been used when calculating the mean the correct mean can be obtained without repeating the whole process using:

$$\text{Correct Mean} = \text{Wrong Mean} + \frac{(\text{Correct Value} - \text{Wrong Value})}{n}$$

Where n is total number of observations.

Example: An average weight of 10 students was calculated to be 65. Later it was discovered that one weight was misread as 40 instead of 80 k.g. Calculate the correct average weight.

Solutions:

$$\text{Correct Mean} = \text{Wrong Mean} + \frac{(\text{Correct Value} - \text{Wrong Value})}{n}$$

$$\text{Correct Mean} = 65 + \frac{(80 - 40)}{10} = 65 + 4 = 69 \text{ k.g.}$$

5. The effect of transforming original series on the mean.
- If a constant k is added/ subtracted to/ from every observation then the new mean will be $\text{the old mean} \pm k$ respectively.
 - If every observations are multiplied by a constant k then the new mean will be $k * \text{old mean}$

Example:

1. The mean of n Tetracycline Capsules $X_1, X_2, \dots, X_n$ are known to be 12 gm. New set of capsules of another drug are obtained by the linear transformation $Y_i = 2X_i - 0.5$ ($i = 1, 2, \dots, n$) then what will be the mean of the new set of capsules

Solutions:

$$\text{New Mean} = 2 * \text{Old Mean} - 0.5 = 2 * 12 - 0.5 = 23.5$$

2. The mean of a set of numbers is 500.
- If 10 is added to each of the numbers in the set, then what will be the mean of the new set?
 - If each of the numbers in the set are multiplied by -5, then what will be the mean of the new set?

Solutions:

$$\text{a). New Mean} = \text{Old Mean} + 10 = 500 + 10 = 510$$

$$\text{b). New Mean} = -5 * \text{Old Mean} = -5 * 500 = -2500$$

Weighted Mean

- ☞ When a proper importance is desired to be given to different data a weighted mean is appropriate.
- ☞ Weights are assigned to each item in proportion to its relative importance.
- ☞ Let $X_1, X_2, \dots, X_n$ be the value of items of a series and $W_1, W_2, \dots, W_n$ their corresponding weights, then the weighted mean denoted $\bar{X}_w$ is defined as:

$$\bar{X}_w = \frac{\sum_{i=1}^n X_i W_i}{\sum_{i=1}^n W_i}$$

Example:

A student obtained the following percentage in an examination:

English 60, Biology 75, Mathematics 63, Physics 59, and chemistry 55. Find the student's weighted arithmetic mean if weights 1, 2, 1, 3, 3 respectively are allotted to the subjects.

Solutions:

$$\bar{X}_w = \frac{\sum_{i=1}^5 X_i W_i}{\sum_{i=1}^5 W_i} = \frac{60*1 + 75*2 + 63*1 + 59*3 + 55*3}{1 + 2 + 1 + 3 + 3} = \frac{615}{10} = 61.5$$

Example: You took 5 courses: a 3 credit stat course and earned an A, a 4 credit Chemistry course and earned a B, a civics and ethics course that was 2 credits and earned an A, a Maths course that was 3 credits and earned a C, and a sociology course that was 3 credits and earned an A. What was your semester GPA for that semester?

Course	Credits (Weight)	Grade (x)	Weight × Value
Stat	3	A (4.0)	12
Chem	4	B (3.0)	12
Civic	2	A (4.0)	8
Maths	3	C (2.0)	6
Sociology	3	A (4.0)	12
Total	15		50

$$\bar{x} = \sum wx / \sum w = 50 / 15 = 3.333$$

Exercise: suppose there are three markets A, B, and C for the sale of a certain item. If 30 units of the items are sold in market A at a unit price of Birr 10, 10 units are sold at a unit price of birr 20 in Market B and 20 units are sold at a unit price of birr 15 in market C. Calculate the average unit price of items. **Answer: 13.33 Birr**

Merits and Demerits of Arithmetic Mean**Merits:**

- It is rigidly defined.
- It is based on all observations.
- It is suitable for further mathematical treatment.
- It is a stable average, i.e. it is not affected by fluctuations of sampling to some extent.
- It is easy to calculate and simple to understand.
- It can be computed for interval and ratio data.

Demerits:

- It is affected by extreme observations.
- It cannot be used in the case of open end classes.

- It can not be determined by the method of inspection.
- It can not be used when dealing with qualitative characteristics, such as intelligence, honesty, beauty.
- It can be a number which does not exist in a series.
- Some times it leads to wrong conclusion if the details of the data from which it is obtained are not available.
- It gives high weight to high extreme values and less weight to low extreme values.
- It is not computed for Nominal or Ordinal data

The Geometric Mean

- ☞ The geometric mean of a set of n observations is the n^{th} root of their product.
- ☞ The geometric mean of $X_1, X_2, X_3 \dots X_n$ is denoted by G.M and given by:

$$G.M = \sqrt[n]{X_1 * X_2 * \dots * X_n}$$

- ☞ Taking the logarithms of both sides

$$\log(G.M) = \log(\sqrt[n]{X_1 * X_2 * \dots * X_n}) = \log(X_1 * X_2 * \dots * X_n)^{\frac{1}{n}}$$

$$\Rightarrow \log(G.M) = \frac{1}{n} \log(X_1 * X_2 * \dots * X_n) = \frac{1}{n} (\log X_1 + \log X_2 + \dots + \log X_n)$$

$$\Rightarrow \log(G.M) = \frac{1}{n} \sum_{i=1}^n \log X_i$$

$\Rightarrow$ The logarithm of the G.M of a set of observations is the arithmetic mean of their logarithms.

$$\Rightarrow G.M = \text{Antilog}\left(\frac{1}{n} \sum_{i=1}^n \log X_i\right)$$

Example:

Find the G.M of the numbers 2, 4, 8.

Solutions:

$$G.M = \sqrt[n]{X_1 * X_2 * \dots * X_n} = \sqrt[3]{2 * 4 * 8} = \sqrt[3]{64} = 4$$

Remark: The Geometric Mean is useful and appropriate for finding averages for ratios, percentages, proportions, indices or growth rates

The Harmonic Mean

The harmonic mean of $X_1, X_2, X_3 \dots X_n$ is denoted by H.M and given by:

$$H.M = \frac{n}{\sum_{i=1}^n \frac{1}{X_i}}, \text{ This is called simple harmonic mean.}$$

In a case of frequency distribution:

$$H.M = \frac{n}{\sum_{i=1}^k \frac{f_i}{X_i}}, \quad n = \sum_{i=1}^k f_i$$

If observations $X_1, X_2, \dots X_n$ have weights $W_1, W_2, \dots W_n$ respectively, then their harmonic mean is given by

$$H.M = \frac{\sum_{i=1}^n W_i}{\sum_{i=1}^n W_i / X_i}, \text{ This is called Weighted Harmonic Mean.}$$

Remark: The Harmonic Mean is useful and appropriate in finding averages to speed, rate and time.

Example: A cyclist pedals from his house to his college at speed of 10 km/hr and back from the college to his house at 15 km/hr. Find the average speed.

Solution: Here the distance is constant

→ The simple H.M is appropriate for this problem.

$$X_1 = 10 \text{ km/hr} \quad X_2 = 15 \text{ km/hr.} \quad H.M = \frac{2}{\frac{1}{10} + \frac{1}{15}} = 12 \text{ km/hr}$$

Example: Suppose you spend \$6 on pills costing 30 cents per dozen, and \$6 on pills costing 20 cents per dozen. What was the average price of the pills you bought?

You spent \$12 on 50 dozen pills, so the average cost is $12/50 = 0.24$, or 24 cents.

This also happens to be the harmonic mean of 20 and 30:

$$\frac{2}{\frac{1}{30} + \frac{1}{20}} = 24$$

For the data set:

1, 2, 3, 4, 5, 6, 7, 8, 9, 10: the Arithmetic mean = 5.50, Geometric Mean = 4.53, Harmonic mean = 3.41 and the root mean square = 6.20

Relations among different means

1. $\bar{x} \geq GM \geq HM$

2. For two observations $\sqrt{\bar{x} * HM} = GM$

3. $\bar{x} = GM = HM$ if all observation have equal magnitude

The Mode

- Mode is a value which occurs most frequently in a set of values
- The mode may not exist and even if it does exist, it may not be unique.
- In case of discrete distribution the value having the maximum frequency is the model value.

Examples:

1. Find the mode of 5, 3, 5, 8, 9

Mode = 5

2. Find the mode of 8, 9, 9, 7, 8, 2, and 5.

It is a bimodal Data: 8 and 9

3. Find the mode of 4, 12, 3, 6, and 7.

No mode for this data.

- The mode of a set of numbers $X_1, X_2, \dots, X_n$ is usually denoted by $\hat{X}$.

Mode for Grouped data

If data are given in the shape of continuous frequency distribution, the mode is defined as:

$$\hat{X} = L_{mo} + w \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right)$$

Where:

$\hat{X}$ = the mode of the distribution

w = the size of the modal class

$\Delta_1 = f_{mo} - f_1$

$\Delta_2 = f_{mo} - f_2$

f_{mo} = frequency of the modal class

f_1 = frequency of the class preceeding the modal class

f_2 = frequency of the class following the modal class

Note: The modal class is a class with the highest frequency.

Example: Following is the distribution of the size of certain farms selected at random from a district. Calculate the mode of the distribution.

Size of farms	No. of farms
5-15	8
15-25	12
25-35	17
35-45	29
45-55	31
55-65	5
65-75	3

Solutions:

45 – 55 is the modal class, since it is a class with the highest frequency.

$L_{mo} = 45$

$w = 10, \Delta_1 = f_{mo} - f_1 = 2$

$\Delta_2 = f_{mo} - f_2 = 26, f_{mo} = 31, f_1 = 29, f_2 = 5$

$\Rightarrow \hat{X} = 45 + 10 \left(\frac{2}{2 + 26} \right) = 45.71$

Merits and Demerits of Mode

Merits:

- It is not affected by extreme observations.
- Easy to calculate and simple to understand.
- It can be calculated for distribution with open end class (if the modal classes are not in the lowest and upper classes)
- It can be determined for all levels of measurement scales/data/

Demerits:

- It is not rigidly defined.
- It is not based on all observations
- It is not suitable for further mathematical treatment.
- It is not stable average, i.e. it is affected by fluctuations of sampling to some extent.
- Often its value is not unique.
- It may not exist for a given data set/ distribution/

Note: being the point of maximum density, mode is especially useful in finding the most popular size in studies relating to marketing, trade, business, and industry. It is the appropriate average to be used to find the ideal size.

The Median

- In a distribution, median is the value of the variable which divides it in to two equal halves.
- In an ordered series of data median is an observation lying exactly in the middle of the series.

It is the middle most value in the sense that the number of values less than the median is equal to the number of values greater than it.

-If $X_1, X_2, \dots, X_n$ be the observations, then the numbers arranged in ascending order will be $X_{[1]}, X_{[2]}, \dots, X_{[n]}$, where $X_{[i]}$ is i^{th} smallest value. $\Rightarrow X_{[1]} < X_{[2]} < \dots < X_{[n]}$

-Median is denoted by $\hat{X}$.

Median for ungrouped data

$$\tilde{X} = \begin{cases} X_{[(n+1)/2]} & , \text{If } n \text{ is odd.} \\ \frac{1}{2}(X_{[n/2]} + X_{[(n/2)+1]}), & \text{If } n \text{ is even} \end{cases}$$

Example: Find the median of the following numbers.

- 6, 5, 2, 8, 9, 4.
- 2, 1, 8, 3, 5, 8.

Solutions:

- First order the data: 2, 4, 5, 6, 8, 9
Here $n=6$

$$\tilde{X} = \frac{1}{2}(X_{[\frac{n}{2}]} + X_{[\frac{n}{2}+1]}) = \frac{1}{2}(X_{[3]} + X_{[4]}) = \frac{1}{2}(5 + 6) = 5.5$$

- Order the data : 1, 2, 3, 5, 8
Here $n=5$

$$\tilde{X} = X_{[\frac{n+1}{2}]} = X_{[3]} = 3$$

Median for grouped data

If data are given in the shape of continuous frequency distribution, the median is defined as:

$$\tilde{X} = L_{med} + \frac{w}{f_{med}} \left(\frac{n}{2} - c \right)$$

Where :

L_{med} = lower class boundary of the median class.

w = the size of the median class

n = total number of observations.

c = the cumulative frequency (less than type) preceeding the median class.

f_{med} = the frequency of the median class.

Remark:

The median class is the class with the smallest cumulative frequency (less than type) greater than or equal to $\frac{n}{2}$.

Example: Find the median of the following distribution.

Class	Frequency
40-44	7
45-49	10
50-54	22
55-59	15
60-64	12
65-69	6
70-74	3

Solutions:

- First find the less than cumulative frequency.
- Identify the median class.
- Find median using formula.

Class	Frequency	Cumu.Freq(less than type)
40-44	7	7
45-49	10	17
50-54	22	39
55-59	15	54
60-64	12	66
65-69	6	72
70-74	3	75

$$\frac{n}{2} = \frac{75}{2} = 37.5$$

39 is the first cumulative frequency to be greater than or equal to 37.5

⇒ 50 – 54 is the median class.

$$L_{med} = 49.5, \quad w = 5, n = 75, \quad c = 17, \quad f_{med} = 22$$

$$\Rightarrow \tilde{X} = L_{med} + \frac{w}{f_{med}} \left(\frac{n}{2} - c \right) = 49.5 + \frac{5}{22} (37.5 - 17) = 54.16$$

Merits and Demerits of Median

Merits:

- Median is a positional average and hence not influenced by extreme observations.
- Can be calculated in the case of open ended classes (the lost and upper class are not median class).
- Median can be located even if the data are incomplete.
- It is unique
- It can be computed for ratio, interval and ordinal data.

Demerits:

- It is not a good representative of data if the number of items is small.
- It is not amenable to further algebraic treatment.
- It is susceptible to sampling fluctuations.
- The ranking or order of magnitudes of data values is tedious for large data set.

Exercise: 2 3 4 4 4 6 8 9 10 11 11

Quantiles

When a distribution is arranged in order of magnitude of items, the median is the value of the middle term. Their measures that depend up on their positions in distribution quartiles, deciles, and percentiles are collectively called quantiles.

Quartiles:

- Quartiles are measures that divide the frequency distribution in to four equal parts.
- The value of the variables corresponding to these divisions are denoted Q_1 , Q_2 , and Q_3 often called the first, the second and the third quartile respectively.
- Q_1 is a value which has 25% items which are less than or equal to it. Similarly Q_2 has 50% items with value less than or equal to it and Q_3 has 75% items whose values are less than or equal to it.
- To find Q_i ($i=1, 2, 3$) we count $\frac{iN}{4}$ of the classes beginning from the lowest class.
- For grouped data: we have the following formula

$$Q_i = L_{Q_i} + \frac{w}{f_{Q_i}} \left(\frac{iN}{4} - c \right) \quad , i = 1, 2, 3$$

Where :

L_{Q_i} = lower class boundary of the quartile class.

w = the size of the quartile class

Re

N = total number of observations.

c = the cumulative frequency (less than type) preceeding the quartile class.

f_{Q_i} = the frequency of the quartile class.

mark:

The quartile class (class containing Q_i) is the class with the smallest cumulative frequency (less than type) greater than or equal to $\frac{iN}{4}$.

Deciles:

- Deciles are measures that divide the frequency distribution in to ten equal parts.
- The values of the variables corresponding to these divisions are denoted $D_1, D_2, \dots, D_9$ often called the first, the second, ..., the ninth decile respectively.
- To find D_i ($i=1, 2, \dots, 9$) we count $\frac{iN}{10}$ of the classes beginning from the lowest class
- For grouped data: we have the following formula

$$D_i = L_{D_i} + \frac{w}{f_{D_i}} \left(\frac{iN}{10} - c \right) \quad , i = 1, 2, \dots, 9$$

Where :

L_{D_i} = lower class boundary of the decile class.

w = the size of the decile class

N = total number of observations.

c = the cumulative frequency (less than type) preceeding the decile class.

f_{D_i} = the frequency of the decile class.

Remark:

The decile class (class containing D_i) is the class with the smallest cumulative frequency (less than type) greater than or equal to $\frac{iN}{10}$.

Percentiles:

- Percentiles are measures that divide the frequency distribution in to hundred equal parts.
- The values of the variables corresponding to these divisions are denoted $P_1, P_2, \dots, P_{99}$ often called the first, the second, ..., the ninety-ninth percentile respectively.
- To find P_i ($i=1, 2, \dots, 99$) we count $\frac{iN}{100}$ of the classes beginning from the lowest class.
- For grouped data: we have the following formula

$$P_i = L_{p_i} + \frac{w}{f_{p_i}} \left(\frac{iN}{100} - c \right) \quad , i = 1, 2, \dots, 99$$

Where:

L_{p_i} = lower class boundary of the percentile class.

w = the size of the percentile class

N = total number of observations.

c = the cumulative frequency (less than type) preceeding the percentile class.

f_{p_i} = the frequency of the percentile class.

Remark:

The percentile class (class containing P_i) is the class with the smallest cumulative frequency (less than type) greater than or equal to $\frac{iN}{100}$.

Example: Considering the following distribution

Calculate:

- All quartiles.
- The 7th decile.
- The 90th percentile.

Values	Frequency
140- 150	17
150- 160	29
160- 170	42
170- 180	72
180- 190	84
190- 200	107
200- 210	49
210- 220	34
220- 230	31
230- 240	16
240- 250	12

Solutions:

- First find the less than cumulative frequency.
- Use the formula to calculate the required quantiles.

Values	Frequency	Cum.Freq(less than type)
140- 150	17	17
150- 160	29	46
160- 170	42	88
170- 180	72	160
180- 190	84	244
190- 200	107	351
200- 210	49	400
210- 220	34	434
220- 230	31	465

230- 240	16	481
240- 250	12	493

a) Quartiles:

i. Q_1

- determine the class containing the first quartile.

$$\frac{N}{4} = 123.25$$

$\Rightarrow 170 - 180$ is the class containing the first quartile.

$$L_{Q_1} = 170, \quad w = 10$$

$$N = 493, \quad c = 88, \quad f_{Q_1} = 72$$

$$\Rightarrow Q_1 = L_{Q_1} + \frac{w}{f_{Q_1}} \left(\frac{N}{4} - c \right) = 170 + \frac{10}{72} (123.25 - 88) = \underline{\underline{174.90}}$$

ii. Q_2

- determine the class containing the second quartile.

$$\frac{2 * N}{4} = 246.5$$

$\Rightarrow 190 - 200$ is the class containing the second quartile.

$$L_{Q_2} = 190, \quad w = 10$$

$$N = 493, \quad c = 244, \quad f_{Q_2} = 107$$

$$\Rightarrow Q_2 = L_{Q_2} + \frac{w}{f_{Q_2}} \left(\frac{2 * N}{4} - c \right) = 190 + \frac{10}{107} (246.5 - 244) = \underline{\underline{190.23}}$$

iii. Q_3

- determine the class containing the third quartile.

$$\frac{3 * N}{4} = 369.75$$

$\Rightarrow 200 - 210$ is the class containing the third quartile.

$$L_{Q_3} = 200, \quad w = 10, N = 493, \quad c = 351, \quad f_{Q_3} = 49$$

$$\Rightarrow Q_3 = L_{Q_3} + \frac{w}{f_{Q_3}} \left(\frac{3 * N}{4} - c \right) = 200 + \frac{10}{49} (369.75 - 351) = \underline{\underline{203.83}}$$

b) D_7

- determine the class containing the 7th decile.

$$\frac{7 * N}{10} = 345.1$$

$\Rightarrow 190 - 200$ is the class containing the seventh decile.

$$L_{D_7} = 190, \quad w = 10$$

$$N = 493, \quad c = 244, \quad f_{D_7} = 107$$

$$\Rightarrow D_7 = L_{D_7} + \frac{w}{f_{D_7}} \left(\frac{7 * N}{10} - c \right) = 190 + \frac{10}{107} (345.1 - 244) = \underline{\underline{199.45}}$$

c) P_{90}

- determine the class containing the 90th percentile.

$$\frac{90 * N}{100} = 443.7$$

$\Rightarrow 220 - 230$ is the class containing the 90th percentile.

$$L_{P_{90}} = 220, \quad w = 10$$

$$N = 493, \quad c = 434, \quad f_{P_{90}} = 3107$$

$$\Rightarrow P_{90} = L_{P_{90}} + \frac{w}{f_{P_{90}}} \left(\frac{90 * N}{100} - c \right) = 220 + \frac{10}{31} (443.7 - 434) = \underline{\underline{223.13}}$$

Mode

2, 2, 3, 3, 4, 4, 4, 4, 5, 6, 7, 8, 8, 8, 9

The mode = **4**

4 is the most frequently occurring value

3.2 Median for values in 3.1

Arrange in ascending order:

2, 2, 3, 3, 4, 4, 4, 4, 5, 6, 7, 8, 8, 8, 9

There are 15 terms.

Since there are an odd number of terms, the median is the middle number.

The median = **4**

Using the formula, the median is located

$$\text{at the } \frac{n+1}{2} \text{ th term} = \frac{15+1}{2} = \frac{16}{2} = 8^{\text{th}} \text{ term}$$

.The 8th term = **4**

3.3 Median

Arrange terms in ascending order:

073, 167, 199, 213, 243, 345, 444, 524, 609, 682

There are 10 terms.

Since there are an even number of terms, the median is the average of the two middle terms: Median = $\frac{243 + 345}{2} = \frac{588}{2} = 294$

Using the formula, the median is located at the $\frac{n+1}{2}$ th term.

$n = 10$ therefore $\frac{10+1}{2} = \frac{11}{2} = 5.5^{\text{th}}$ term.

2 2

The median is located halfway between the 5th and 6th terms.
 5th term = 243 6th term = 345

Halfway between 243 and 345 is the median = **294**

3.4 Mean

17.3

44.5

$$\mu = \Sigma x / N = 333.6 / 8 = \mathbf{41.7}$$

31.6

40.0

52.8

$$\bar{x} = \Sigma x / n = 333.6 / 8 = \mathbf{41.7}$$

38.8

30.1

78.5

$$\Sigma x = 333.6$$

(It is not stated in the problem whether the data represent as population or a sample).

3.5 Mean

7

-2

5

$$\mu = \Sigma x / N = -12 / 12 = \mathbf{-1}$$

9

0

-3

-6

$$\bar{x} = \Sigma x / n = -12 / 12 = \mathbf{-1}$$

-7

-4

-5

2

-8

$$\Sigma x = -12$$

(It is not stated in the problem whether the data represent a population or a sample).

3.6 Rearranging the data into ascending order:

11, 13, 16, 17, 18, 19, 20, 25, 27, 28, 29, 30, 32, 33, 34

$$i = \frac{35}{100}(15) = 5.25, P_{35} \text{ is located at the } 5 + 1 = 6^{\text{th}} \text{ term, } P_{35} = \mathbf{19} \quad i = \frac{55}{100}(15) = 8.25, P_{55}$$

is located at the $8 + 1 = 9^{\text{th}}$ term. $P_{55} = \mathbf{27}$. $Q_1 = P_{25}$

$$i = \frac{25}{100}(15) = 3.75 \quad Q_1 = P_{25} \text{ is located at the } 3 + 1 = 4^{\text{th}} \text{ term. } Q_1 = \mathbf{17}$$

$Q_2 = \text{Median}$

The median is located at the $\left(\frac{15+1}{2}\right)^{\text{th}} = 8^{\text{th}} \text{ term} \quad Q_2 = \mathbf{25} \quad Q_3 = P_{75}$

$$i = \frac{75}{100}(15) = 11.25. Q_3 = P_{75} \text{ is located at the } 11 + 1 = 12^{\text{th}} \text{ term. } Q_3 = \mathbf{30}$$

3.7 Rearranging the data in ascending order:

80, 94, 97, 105, 107, 112, 116, 116, 118, 119, 120, 127,

128, 138, 138, 139, 142, 143, 144, 145, 150, 162, 171, 172

$$n = 24$$

$$i = \frac{20}{100}(24) = 4.8 \text{ } P_{20} \text{ is located at the } 4 + 1 = 5^{\text{th}} \text{ term. } P_{20} = \mathbf{107}$$

$$i = \frac{47}{100}(24) = 11.28 \text{ } P_{47} \text{ is located at the } 11 + 1 = 12^{\text{th}} \text{ term. } P_{47} = \mathbf{127}$$

$$i = \frac{83}{100}(24) = 19.92 \text{ } P_{83} \text{ is located at the } 19 + 1 = 20^{\text{th}} \text{ term. } P_{83} = \mathbf{145}$$

$$Q_1 = P_{25}$$

$$i = \frac{25}{100}(24) = 6 \text{ } Q_1 \text{ is located at the } 6.5^{\text{th}} \text{ term}$$

$$Q_1 = (112 + 116)/2 = \mathbf{114} \text{ } Q_2 = \text{Median}$$

$$\text{The median is located at the: } \left(\frac{24+1}{2} \right)^{\text{th}} = 12.5^{\text{th}} \text{ term}$$

$$Q_2 = (127 + 128)/2 = \mathbf{127.5} \text{ } Q_3 = P_{75}$$

$$i = \frac{75}{100}(24) = 18 \text{ } Q_3 \text{ is located at the } 18.5^{\text{th}} \text{ term. } Q_3 = (143 + 144)/2 = \mathbf{143.5}$$

$$3.8 \text{ Mean} = \frac{\sum x}{N} = \frac{43,063}{15} = 2,870.87$$

$$\text{The median is located at the } \left(\frac{15+1}{2} \right)^{\text{th}} = 8^{\text{th}} \text{ term. Median} = \mathbf{2,264} \text{ } Q_2 = \text{Median} = 2,264$$

$$i = \frac{63}{100}(15) = 9.45 \text{ } P_{63} \text{ is located at the } 9 + 1 = 10^{\text{th}} \text{ term. } P_{63} = \mathbf{2,646}$$

$$3.9 \text{ The median is located at the } \left(\frac{10+1}{2} \right)^{\text{th}} = 5.5^{\text{th}} \text{ position}$$

$$\text{The median} = (595 + 653)/2 = \mathbf{624} \text{ } Q_3 = P_{75}: i = \frac{75}{100}(10) = 7.5$$

$$P_{75} \text{ is located at the } 7+1 = 8^{\text{th}} \text{ term. } Q_3 = \mathbf{751}$$

$$\text{For } P_{20}: i = \frac{20}{100}(10) = 2 \text{ } P_{20} \text{ is located at the } 2.5^{\text{th}} \text{ term. } P_{20} = (483 + 489)/2 = \mathbf{486}$$

$$\text{For } P_{60}: i = \frac{60}{100}(10) = 6 \text{ } P_{60} \text{ is located at the } 6.5^{\text{th}} \text{ term. } P_{60} = (653 + 701)/2 = \mathbf{677}$$

$$\text{For } P_{80}: i = \frac{80}{100}(10) = 8 \text{ } P_{80} \text{ is located at the } 8.5^{\text{th}} \text{ term. } P_{80} = (751 + 800)/2 = \mathbf{775.5}$$

$$\text{For } P_{93}: i = \frac{93}{100}(10) = 9.3 \text{ } P_{93} \text{ is located at the } 9+1 = 10^{\text{th}} \text{ term. } P_{93} = \mathbf{1096}$$

$$3.10 \text{ } n = 17$$

$$\text{Mean} = \frac{\sum x}{N} = \frac{61}{17} = 3.59 \text{ } \text{The median is located at the } \left(\frac{17+1}{2} \right)^{\text{th}} = 9^{\text{th}} \text{ term}$$

Median = **4**. Mode = **4**. $Q_3 = P_{75}$: $i = \frac{75}{100}(17) = 12.75$. Q_3 is located at the 13th term. $Q_3 = \mathbf{4}$

P_{11} : $i = \frac{11}{100}(17) = 1.87$. P_{11} is located at the 2nd term. $P_{11} = \mathbf{1}$ P_{35} : $i = \frac{35}{100}(17) = 5.95$

P_{35} is located at the 6th term. $P_{35} = \mathbf{3}$

P_{58} : $i = \frac{58}{100}(17) = 9.86$. P_{58} is located at the 10th term. $P_{58} = \mathbf{4}$

P_{67} : $i = \frac{67}{100}(17) = 11.39$. P_{67} is located at the 12th term. $P_{67} = \mathbf{4}$